

Fourth Workshop on Refactoring Tools (WRT 2011)

Danny Dig
University of Illinois at Urbana-Champaign
Urbana, IL 61801 USA
dig@illinois.edu

Don Batory
University of Texas at Austin
Austin, TX 78712 USA
batory@cs.utexas.edu

ABSTRACT

Refactoring is the process of applying behavior-preserving transformations to a program with the objective of improving the program's design. A specific refactoring is identified by a name (e.g., Extract Method), a set of preconditions, and a set of transformations that need to be performed.

Tool support for refactoring is essential because checking the preconditions of refactoring often requires nontrivial program analysis, and applying transformations may affect many locations throughout a program. In recent years, the emergence of light-weight programming methodologies such as Extreme Programming has generated a great amount of interest in refactoring, and refactoring support has become a required feature in today's IDEs.

This workshop is a continuation of a series of previous workshops (ECOOP 2007, OOPSLA 2008 and 2009 – see <http://refactoring.info/WRT>) where researchers and developers of refactoring tools can meet and discuss recent ideas and work, and view tool demonstrations.

Categories and Subject Descriptors

D.2.3 [Coding Tools and Techniques]: Program Editors; D.2.7 [Distribution, Maintenance, and Enhancement]: Restructuring, reverse engineering, and reengineering

General Terms

Algorithms, Design, Management

Keywords

Refactoring, Program Analysis, Program Manipulation, Transformation

1. GOALS

While there is a great deal of interest in developing tool support for refactoring, researchers and tool vendors rarely

work together. This forum fosters the transfer of ideas and expertise both ways: researchers can show the state-of-the-art analyses they are using in developing tool support for refactoring, tool vendors can offer valuable insights on the challenges of scaling such analyses to realistic applications.

By bringing together researchers and tool vendors, we can shorten the time to transfer ideas into production systems. In addition, by making researchers aware of what others are working on, the potential for reinventing the wheel is reduced and the potential for creative collaboration is enhanced.

2. TOPICS OF INTEREST

Topics that at the forefront of refactoring include:

- refactoring engines
- improving the usability of existing refactoring engines
- empirical studies about refactoring practice
- tool performance
- efficient representation of source models
- program analyses for refactoring tools
- tools for detecting applied refactorings
- tools for suggesting refactorings (e.g., using code-smell detection)
- testing and verification of refactoring tools
- language-independent analysis frameworks and analytical representations
- language-independent transformation frameworks
- language-independent refactoring tools
- refactoring tools for non-OO languages (e.g., functional languages, MDE models and metamodels, legacy languages)
- composition and scripting of refactorings
- medium- and large-scale refactorings (e.g. package-level, component-level)
- refactoring for concurrency and parallelism

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The workshop schedule includes a plenary session for the presentation of selected papers, tool demonstrations, and formal and informal discussions.

Accepted papers will appear on the workshop's home page; the proceedings will be published by ACM.

Participation is open to those who have submitted papers as well as to others who have an interest in refactoring tools.

3. ORGANIZERS

- Danny Dig, University of Illinois at Urbana-Champaign, USA
- Don Batory, University of Texas at Austin, USA

4. PROGRAM COMMITTEE

- Ira Baxter, Semantic Designs, USA
- Jan Becicka, Oracle/NetBeans, Czech Republic
- Bob Fuhrer, IBM TJ Watson, USA
- Bill Griswold, University of California - San Diego, USA
- Ralph Johnson, University of Illinois, USA

- Joshua Kerievsky, Industrial Logic, USA
- Miryung Kim, University of Texas at Austin, USA
- Radu Marinescu, Politechnics University of Timisora, Romania
- Emerson Murphy-Hill, North Carolina State University, USA
- David Notkin, University of Washington, USA
- Bill Opdyke, Motorola, USA
- Kevin Pilch-Bisson, Microsoft/ Visual Studio, USA
- Max Schaefer, Oxford University Computing Laboratory, UK
- Frank Tip, IBM TJ Watson, USA
- Jan Wloka, IBM Zurich/ Eclipse, Switzerland

5. WORKSHOP HOME PAGE

The workshop home page can be found here:

<http://refactoring.info/WRT11>